Chapter 15

Prospects

Centuries from now, superstring theory, or its evolution within M-theory, may have developed so far beyond our current formulation that it might be unrecognizable even to today's leading researchers. As we continue to seek the ultimate theory, we may well find that string theory is but one of many pivotal steps on a path toward a far grander conception of the cosmos—a conception that involves ideas that differ radically from anything we have previously encountered. The history of science teaches us that each time we think that we have it all figured out, nature has a radical surprise in store for us that requires significant and sometimes drastic changes in how we think the world works. Then again, in a bit of brash posturing, we can also imagine, as others before us have perhaps naively done, that we are living through a landmark period in humanity's history in which the search for the ultimate laws of the universe will finally draw to a close. As Edward Witten has said,

> I feel that we are so close with string theory that—in my moments of greatest optimism—I imagine that any day, the final form of the theory might drop out of the sky and land in someone's lap. But more realistically, I feel that we are now in the process of constructing a much deeper theory than anything we have had before and that well into the twenty-first century, when I am too old to have any useful thoughts on the subject, younger physicists will have to decide whether we have in fact found the final theory.1

Although we are still feeling the aftershocks of the second superstring revolution and absorbing the panoply of new insights that it has engendered, most string theorists agree that it will likely take a third and maybe a fourth such theoretical upheaval before the full power of string theory is unleashed and its possible role as the final theory assessed. As we have seen, string theory has already painted a remarkable new picture of how the universe works, but there are significant hurdles and loose ends that will no doubt be the primary focus of string theorists in the twenty-first century. And so, in this last chapter we will not be able to finish telling the story of humanity's search for the deepest laws of the universe, because the search